ANXIETY

INTRODUCTION

Anxiety is an emotional state commonly caused by the perception of real or perceived danger that threatens the security of an individual. It allows a person to prepare for or react to environmental changes.

Individual experience persistent, severe anxiety symptoms and possess irrational fears that significantly impair normal daily functioning.

ETIOLOGY

The differential diagnosis of anxiety disorders includes medical and psychiatric illnesses and certain drugs.

MEDICAL DISEASES ASSOCIATED WITH ANXIETY

- **Cardiovascular** - Angina, arrhythmias, cardiomyopathy, congestive heart failure, hypertension, ischemic heart disease, mitral valve prolapse, myocardial infarction
- **Endocrine and metabolic** - Cushing disease, diabetes, hyperparathyroidism, hyperthyroidism, hypothyroidism, hypoglycemia, hyponatremia, hyperkalemia, pheochromocytoma, vitamin B12 or folate deficiencies
- **Gastrointestinal-** Crohn's disease, irritable bowel syndrome, ulcerative colitis, peptic ulcer disease
- **Neurologic-** Migraine, seizures, stroke, neoplasms, poor pain control
- **Respiratory system-** Asthma, chronic obstructive pulmonary disease, pulmonary embolism, pneumonia
- **Others-** Anemias, cancer, systemic lupus erythematosus, vestibular dysfunction

PSYCHIATRIC DISEASES ASSOCIATED WITH ANXIETY

Anxiety symptoms are extremely common in patients with mood disorders, schizophrenia, dementia, and substance use disorders.

DRUG INDUCED ANXIETY

- Anticonvulsants: Carbamazepine, phenytoin
- Antidepressants: Bupropion, selective serotonin reuptake inhibitors, serotonin–norepinephrine reuptake inhibitors
- Antihypertensives: Clonidine, felodipine
- Antibiotics: Quinolones, isoniazid
- Bronchodilators: Albuterol, theophylline
- Corticosteroids: Prednisone
- Dopamine agonists: Amantadine, levodopa
- Herbals: Ma huang, ginseng, ephedra
- Illicit substances: Ecstasy, marijuana
- Nonsteroidal anti-inflammatory drugs: Ibuprofen, indomethacin
- Stimulants: Amphetamines, caffeine, cocaine, methylphenidate, nicotine
- Sympathomimetics: Pseudoephedrine, phenylephrine
- Thyroid hormones: Levothyroxine
- Toxicity: Anticholinergics, antihistamines, digoxin

TYPES OF ANXIETY-

1. **Panic disorder**
2. **Generalised anxiety disorder**
3. **Post traumatic stress disorder**
4. **Social anxiety disorder**
5. **Simple phobia**
6. **Obsessive compulsive disorder**

CLINICAL PRESENTATION

GENERALISED ANXIETY DISORDER

Psychological and Cognitive Symptoms-	**Physical Symptoms**
• Excessive anxiety • Worries that are difficult to control • Feeling keyed up or on edge • Trouble concentrating or mind going blank	• Restlessness • Fatigue • Muscle tension • Sleep disturbance • Irritability

PANIC DISORDER

Psychological Symptoms	**Physical Symptoms**
• Depersonalization • Derealization • Fear of losing control, going crazy, or dying	• Abdominal distress • Chest pain or discomfort • Chills • Dizziness or light-headedness • Feeling of choking • Hot flushes • Palpitations • Nausea • Paresthesias • Shortness of breath • Sweating • Tachycardia • Trembling or shaking

SOCIAL ANXIETY DISORDER

Fears of Being	Some Feared Situations	Physical Symptoms	Types
• Scrutinized by others • Embarrassed • Humiliated	• Eating or writing in front of others • Interacting with authority figures • Speaking in public • Talking with strangers • Use of public toilets	• Blushing • "Butterflies in the stomach" • Diarrhea • Sweating • Tachycardia • Trembling	• Generalized: fear and avoidance extend to a wide range of social situations • Nongeneralized: fear limited to one or two situations

POST TRAUMATIC STRESS DISORDER

Intrusion Symptoms	Avoidance Symptoms	Persistent Negative Alterations in Thinking and Mood	Hyperarousal Symptoms
• Recurrent, intrusive distressing memories of the trauma • Recurrent, disturbing dreams of the event • Feeling that the traumatic event is recurring (eg, dissociative flashbacks) • Physiologic reaction to or psychological distress from reminders of the trauma	• Avoidance of conversations, thoughts, or feelings about the trauma • Avoidance of people, places, or activities that are reminders of the event	• Inability to recall an important aspect of the trauma • Anhedonia •Estrangement from others • Restricted affect • Negative beliefs about oneself • Distorted beliefs causing one to blame others or themselves for the trauma • Negative mood state	• Decreased concentration • Easily startled • Self-destructive behavior • Hypervigilance • Insomnia • Irritability or anger outbursts

OBSESSIVE-COMPULSIVE DISORDER

Obsessions	Compulsions
• Repetitive thoughts (eg, feeling contaminated by germs, fears of harming others) • Repetitive images (eg, recurrent sexually explicit pictures) • Repetitive urges (eg, need for symmetry or putting things in specific order)	• Repetitive activities (eg, hand washing, need to ask, need to confess) • Repetitive mental acts (eg, counting excessively, repeating words silently, praying)

TREATMENT

TABLE 87-3 Drug Choices for Anxiety Disorders

Anxiety Disorder	First-Line Drugs	Second-Line Drugs	Alternatives
Generalized anxiety disorder	Duloxetine Escitalopram Paroxetine Sertraline Venlafaxine XR	Benzodiazepines Buspirone Imipramine Pregabalin	Hydroxyzine Quetiapine
Panic disorder	SSRIs Venlafaxine XR	Alprazolam Citalopram Clomipramine Clonazepam Imipramine	Phenelzine
Social anxiety disorder	Escitalopram Fluvoxamine CR Paroxetine Sertraline Venlafaxine XR	Clonazepam Citalopram	Gabapentin Phenelzine Pregabalin

CR, controlled-release; SSRI, selective serotonin reuptake inhibitor; XR, extended-release.

GENERALISED ANXIETY DISORDER

The goals of therapy in the acute management of GAD are to reduce the severity and duration of the anxiety symptoms and to improve overall functioning. The long-term goal in GAD is remission with minimal or no anxiety symptoms, no functional impairment, and increased QOL.

NON-PHARMACOLOGICAL TREATMENT

- Nonpharmacologic treatment modalities in GAD include psychoeducation, short-term counseling, stress management, psychotherapy, mindfulness-based therapy, or exercise. Psychoeducation includes information on the etiology and management of GAD
- Patients with anxiety should be instructed to avoid caffeine, nicotine, nonprescription stimulants, diet pills, and excessive use of alcohol

PHARMACOLOGICAL

Benzodiazepine Antianxiety Agents		
DRUG	APPROVED DOSAGE RANGE (mg/day)	MAXIMUM DOSAGE FOR GERIATRIC PATIENTS (mg/day)
Alprazolam	0.75–4	2
Clonazepam	1-4	3
Diazepam	2-40	20
Lorazepam	0.5-10	3

Nonbenzodiazepine Antianxiety Agents for Generalized Anxiety Disorder

DRUG		INITIAL DOSE (mg/day)	USUAL RANGE (mg/day)
Antidepressant	Duloxetine	30 or 60	60-120
	Escitalopram	10	10-20
	Sertraline	50	50-200
	Imipramine	50	75-200
Azapirone	Buspirone	7.5 BD	15-602
Diphenylmethane	Hydroxyzine	25 or 50 mg four times daily	200-400
Anticonvulsant	Pregabalin	50 mg TDS	150-600
Atypical antipsychotic	Quetiapine XR	50 mg HS	150-300

ADRS-

- **Selective Serotonin Reuptake Inhibitors and Serotonin–Norepinephrine Reuptake Inhibitors -** Jitterness syndrome, suicidality, nausea, diarrhea, headache, weight gain
- **Tricyclic Antidepressants-** Jitterness syndrome, suicidality, weight gain, sedation, sexual dysfunction.
- **Benzodiazepines-** drowsiness, fatigue, withdrawal syndrome
- **Buspirone-**nausea, drowsiness, abdominal pain
- **Pregabalin-** dizziness, somnolence, weight gain
- **Quetiapine-** sedation, metabolic syndrome, akathisia

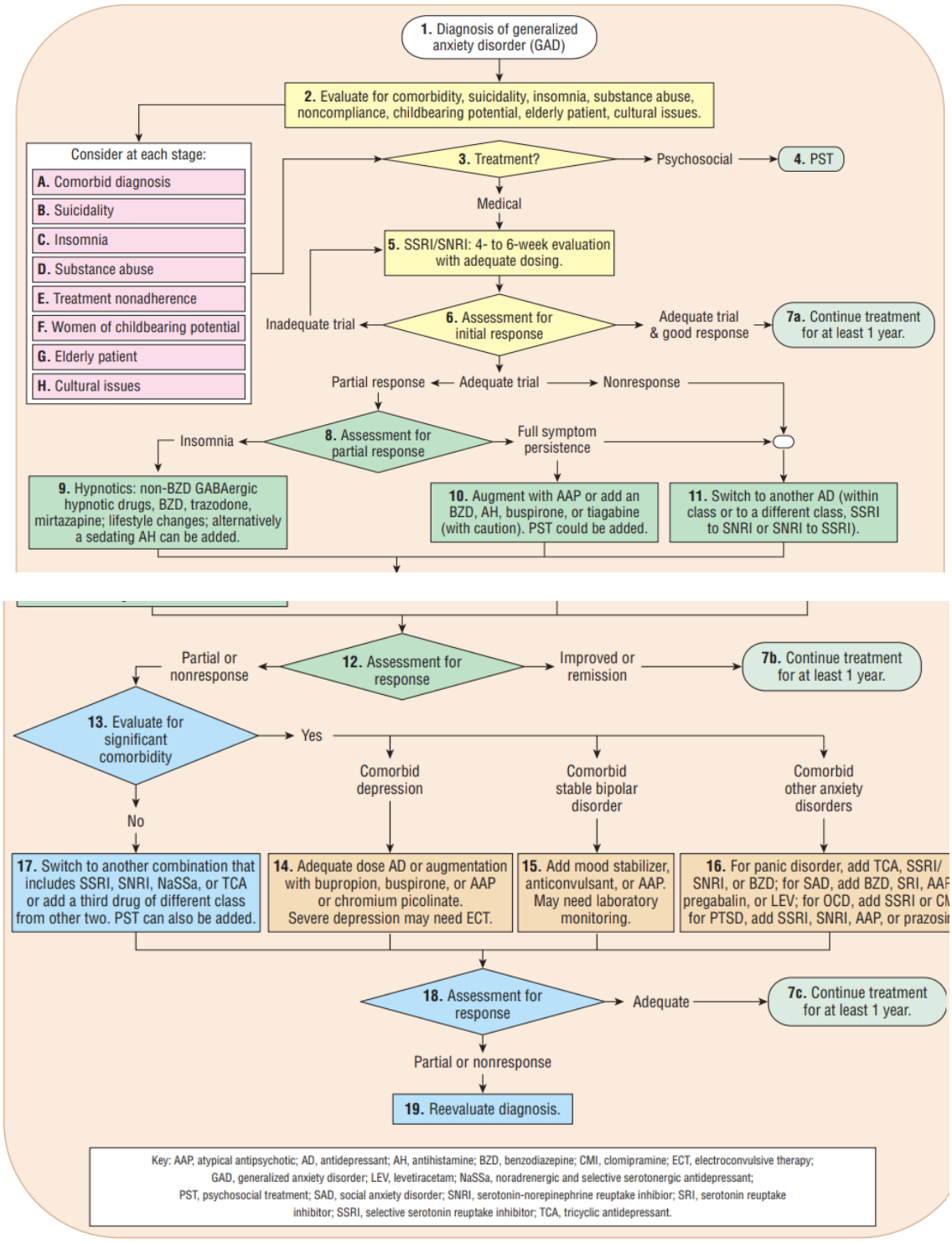


FIGURE 87-1 International Psychopharmacology Algorithm Project (IPAP) generalized anxiety disorder (GAD) algorithm flowchart.

PANIC DISORDER

GOAL OF THERAPY

The goals are complete resolution of panic attacks, marked reduction in anticipatory anxiety, elimination of phobic avoidance, and resumption of normal activities.

PHARMACOTHERAPY

CLASS		STARTING DOSE	ANTIPANIC DOSE RANGE(mg)
SSRIs	Citalopram	10 mg/day	20-40
	Escitalopram	5 mg/day	10-20
	Sertraline	25 mg/day	50-200
	Fluoxetine	5 mg/day	10-30
SNRI	Venlafaxine XR	37.5 mg/day	75-225
Benzodiazepines	Alprazolam	0.25 mg three times a day	4-10
	Clonazepam	0.25 mg once or twice per day	1-4
	Diazepam	2–5 mg three times a day	5-20
	Lorazepam	0.5–1 mg three times a day	2-8
TCA	Imipramine	10 mg/day	75-250
MAO I	Phenelzine	15 mg/day	45-90

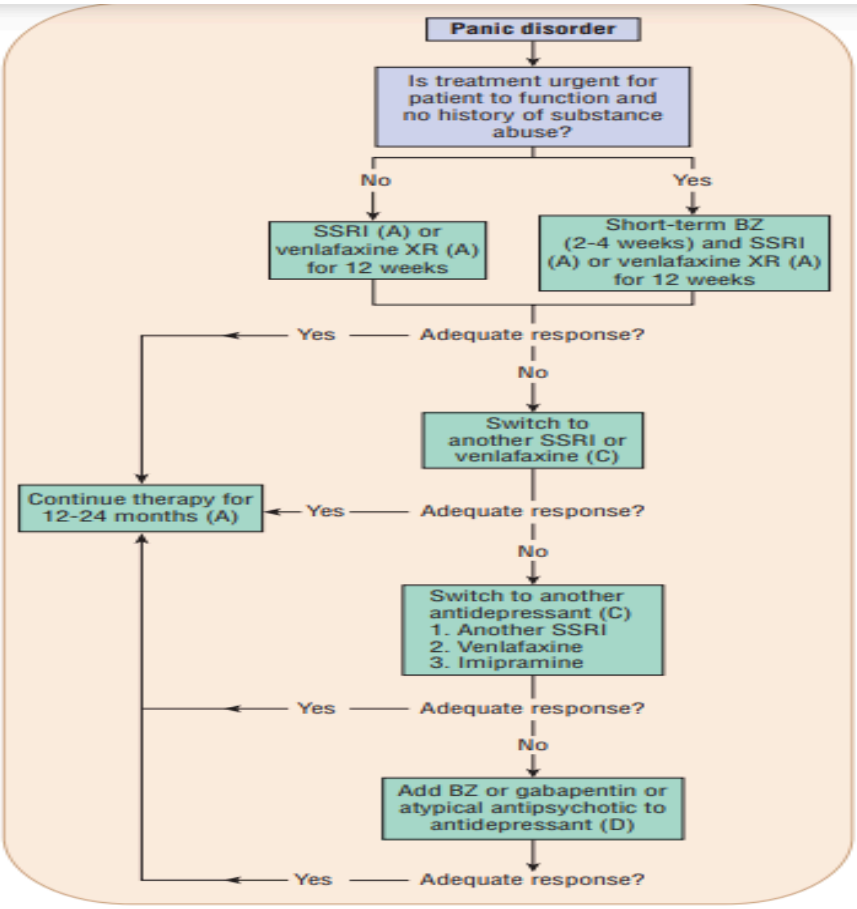


FIGURE 87-2 Algorithm for the pharmacotherapy of panic disorder. Strength of recommendations: A, directly based on

SOCIAL ANXIETY DISOREDER

GOAL OF THERAPY

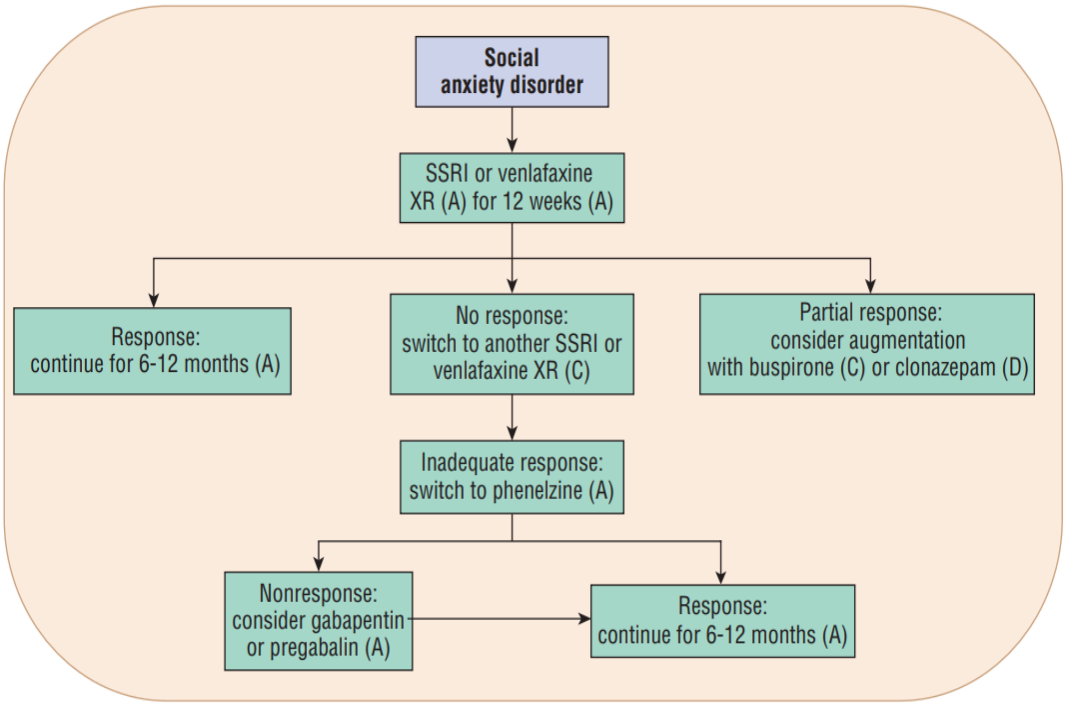
The goals are to reduce the physiologic symptoms and phobic avoidance, increase participation in desired social activities, and improve quality of life.

NONPHARMACOLOGICAL THERAPY

- Patients should be educated about SAD and support groups.
- Self-help group programs that focus on effective communication can benefit people with anxiety involving public speaking.
- In general, CBT consists of exposure therapy, cognitive restructuring, relaxation training techniques, and social skills training.

PHARMACOTHERAPY

CLASS		STARTING DOSE	ANTIPANIC DOSE RANGE(mg)
SSRIs	Citalopram	20 mg/day	20-40
	Escitalopram	5 mg/day	10-20
	Sertraline	25–50 mg/day	50-200
	Paroxetine CR	12.5 mg/day	12.5-37.5
SNRI	Venlafaxine XR	75 mg/day	75-225
Benzodiazepines	Clonazepam	0.25mg/day1-4	
MAO I	Phenelzine	15 mg/day	45-90
Alternative agents	Buspirone	10 mg twice per day	45-60
	Gabapentin	100 mg three times a day	900-3600



POST TRAUMATIC STRESS DISORDER

GOAL OF THERAPY

The goals are to decrease core symptoms, disability, and comorbidity and improve quality of life.

NON-PHARMACOLOGICAL THERAPY

Psychotherapy can be used when a patient suffers from mild symptoms, in patients who prefer not to use medications, or in conjunction with medication in patients with severe symptoms in an effort to improve response.

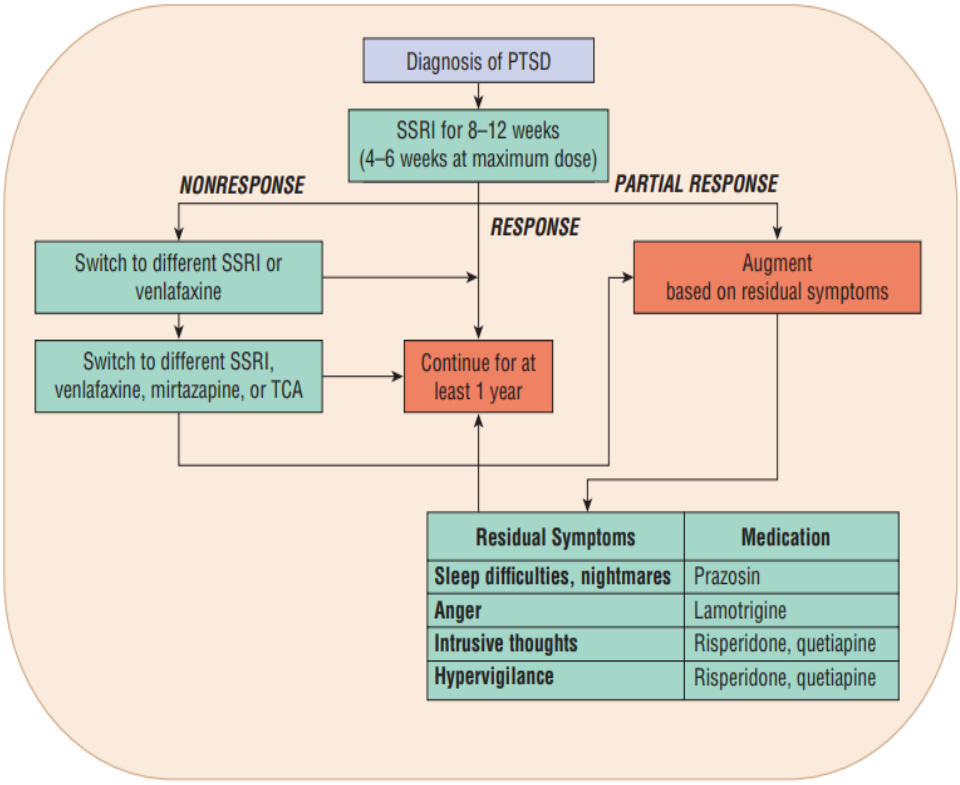
PHARMACOTHERAPY

DRUG		INITIAL DOSE	USUSAL RANGE (mg/day)
SSRI	Fluoxetine	10 mg daily	10-80
	Paroxetine	20-40	10-20 mg daily
	Sertraline	25 mg daily	50-100
Alternative	Amitriptyline	25 or 50 mg daily	75-200
	Imipramine	25 or 50 mg daily	75-200

TABLE 88-2 Summary of Key Points in Treatment Guidelines for PTSD

Recommendations	Level of Evidence	Comments
First-Line Treatments		
SSRIs: Fluoxetine, paroxetine, sertraline SNRIs: Venlafaxine	I I	At 4 weeks if there is partial response, continue for another 4 weeks. At 8 weeks, if no improvement, increase dose to maximum tolerated or switch to another first-line treatment
Second-Line Treatments		
TCAs: Amitriptyline, imipramine	II	The risk of adverse effects and potential for fatalities in a TCA overdose are higher than with SSRIs or SNRIs
Other: Mirtazapine	II	
Augmentation with prazosin for sleep/nightmares	II	Not recommended in the VA guidelines ***for use in veteran population***
Third-Line treatments		
Augmentation with anticonvulsants or atypical antipsychotics	IV	The VA guidelines recommend against using risperidone as an augmenting agent secondary to metabolic adverse effects. There is insufficient evidence to support routine use of other atypical antipsychotics or anticonvulsants.
MAOIs: Phenelzine	III	Potential adverse effects and dietary restrictions limit use.
Other: Nefazodone	II	Tolerability and risk for hepatotoxicity limit use.

PTSD, posttraumatic stress disorder; SNRIs, serotonin–norepinephrine reuptake inhibitors; SSRIs, selective serotonin reuptake inhibitors; TCA, tricyclic antidepressant; VA, Veterans Affairs.



OBSESSIVE COMPULSIVE DISORDER

GOAL OF THERAPY-

Major goals of therapy for OCD include reduction in the frequency and severity of obsessive thoughts and time spent performing compulsive acts.

NON-PHARMACOLOGICAL THERAPY-

- Cognitive Behavioural therapy (CBT) with behavioral techniques (eg, exposure and response prevention [ERP]) is the most common initial nonpharmacologic treatment of choice in OCD and is largely considered to be more efficacious than pharmacotherapy.
- CBT is preferred for motivated patients, particularly children and adolescents, with both mild OCD symptoms and psychiatric comorbidities, and in those with a desire to avoid medications
- Other nonpharmacologic options include neuromodulatory approaches (eg, deep brain stimulation [DBS], TMS, and electroconvulsive therapy [ECT]) and ablative neurosurgery for severely symptomatic patients who have not achieved sustained response to standard of care interventions

PHARMACOTHERAPY

First-Line Treatments	**Second-Line Treatments**	**Third-Line Treatments**	**Maintenance and Discontinuation Phase**
CBT alone SSRI alone CBT+ SSRIs	Switch to another SSRI or clomipramine I Augmentation with antipsychotic-Aripiprazole and risperidone are preferred second-generation antipsychotics for OCD treatment Switch to venlafaxine	Switch to another antipsychotic augmenting agent Switch to duloxetine or mirtazapine Augmentation of SSRI with clomipramine-Citalopram, escitalopram, and sertraline have less risk to cause clinically significant drug–drug interactions with clomipramine	After 1-2 years, gradual taper over several months Periodic CBT booster sessions for 3-6 months

SSRIs and clomipramine are the main antidepressants used in OCD treatment, as current evidence indicates that 5-HT is important for the antiobsessional effects of these medications

DRUG	**INITIAL DOSE**	**USUAL RANGE**	**COMMENTS**
Citalopram	20 mg daily	20-40 mg daily	Maximum dose is 40 mg in adults daily to prevent QTc prolongation; maximum dose of 20 mg daily in elderly patients, CYP2C19 poor metabolizers, or use with concurrent moderate-to-strong CYP2C19 inhibitors (eg, cimetidine, omeprazole)
Clomipramine	25 mg daily	100-250 mg daily	Doses up to 300 mg daily may be needed in some patients. Steady state trough plasma levels (clomipramine and desmethylclomipramine) should be
Escitalopram	10 mg daily	10-20 mg daily	Doses up to 40 mg may be needed in some patients
Fluoxetine	20 mg daily	40-60 mg daily	Doses of 80 mg or higher may be needed in some patients

Drug	Adverse Drug Reaction
Clomipramine	Dry mouth, constipation, nausea, dyspepsia, anorexia, somnolence, tremors, dizziness, nervousness
	Seizures
	Orthostatic hypotension, tachycardia, ECG changes
	Suicidality
	Agranulocytosis, leukopenia
	Weight gain
SSRIs	Nausea, vomiting, diarrhea, sexual dysfunction, headache, insomnia
	Anxiety and agitation
	Discontinuation syndrome
	Suicidality
	QTc prolongation

AGORPHOBIA- it is marked as fear or anxiety about being in at least 2 situation in which escape might be difficult or where help might not be available in the event developing panic like symptoms.

QUESTIONS-

Explain sign and symptoms and treatment of OCD (5)

Write s short note on PTSD (2)

Explain sign and symptoms and treatment of social anxiety disorder (5)

Enlist symptoms of OCD (2)

Write a short note on generalized anxiety disorder (2)

Explain clinical presentation, Pathophysiology,diagnosis and pharmacotherapy of anxiety disorder (15)

Define OCD. Ann give its clinical importance (2)

Define and discuss OCD(5)

Sign and symptoms of GAD(2)

Enlist risk factors of anxiety disorder (2)

Give points to be considered for choice of anxiolytic medication(2)

Write a note on OCD (2)

Clinical presentation of OCD(2)